

rep_newseq.py - new spike sequences (S1/S2 of YP_009724390.1) among nr proteins created in the last 90 days
measured 2026-10-01 on dell; runtimes in red; total 40.8 s (cold: 27.7 + 13.1) / 13.0 s (pre-selection cached)

query: examples/sars2_spike_s1s2_query.fa
S1 = 14-685 (672 aa), S2 = 686-1273 (588 aa)

python3 ~/rep_newseq.py --days 90

flock ~/.nr_dmnd.lock SHARED
(no append while the .dmnd is read)
0.0 s

dmnd_preselect.preselect(new_days=90) 27.7 s cold, 0.0 s cache hit

sub-.dmnd for (criteria, dmnd_hash, day)
already in ~/diamond/preselect?

no

MariaDB: blast_seq_ncbi.createdate >= today - 90 d
(idx_createdate)
3.5 s -> 2,543,666 OIDs

read records from nr_full.dmnd (489 GB)
via its offset table (dmnd_getseq)
20.2 s -> 1,084,065,391 letters

diamond makedb -> sub-.dmnd (1.17 GB)
3.9 s

yes

diamond blastp --very-sensitive, 1 reference block (--block-size 2) 13.0-13.1 s, peak RSS 2.8 GB

algorithm chosen by DIAMOND: query-indexed
(2 short queries vs. 1.08 G letters)
load reference 0.8 s

build reference histograms
4.8 s

16 seed shapes x (reference seed array
+ hash join + search)
~7.4 s

compute alignments (E <= 0.001, all targets,
--dbsize = letters of the full nr)
0.1 s

--ultra-sensitive (64 shapes): 41-46 s, 1.9 GB
histograms 18.6 s + 64 shapes ~21 s
hits byte-identical to --very-sensitive
--index-chunks 1: no effect (query-indexed)

MariaDB metadata per hit accession:
blast_acc + blast_seq + blast_taxon + blast_seq_ncbi
< 0.1 s (510 accessions)

CSV ~/rep_newseq_YYYY-MM-DD.csv (header: query, DB, parameters, runtimes)
+ summary on stdout

found 2026-10-01: 1,016 hits = S1 506 + S2 510
510 distinct subject sequences, all created 2026-09
506 SARS-CoV-2; 4 others in S2 only:
Infectious bronchitis virus (2), Alphacoronavirus sp. (1), Canine coronavirus (1)